

SolarRuns: a flare-triggered Moreton wave with the SurfaceCode MHD solver

Adopted physical parameters, driver calibration, and references

SurfaceCode — SolarRuns

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1 Purpose and scenario

This document records the physical parameter values (with sources) adopted for the **SurfaceCode/SolarRuns** simulation of a *Moreton wave*: the arc-shaped $\text{H}\alpha$ front that races away from large flares at 500–2000 km s^{-1} [1, 2, 7]. Following Uchida [3], the front is modelled as the chromospheric footprint of a *coronal fast-mode shock*: a flare-driven blast propagating through the quiet corona whose lower flank refracts downward onto the chromosphere (the fast speed increases with height through the low corona, bending wavefronts back toward the surface). The simulation solves compressible ideal MHD in a 2D (x, z) vertical slice spanning photosphere $\rightarrow$ chromosphere $\rightarrow$ transition region $\rightarrow$ corona, with the flare modelled as an impulsive-phase pressure pulse in the low corona — the “blast-wave” scenario [11, 9], implemented essentially as in the closest published analogue, the FLASH Moreton-wave simulations of Krause et al. [12]. The solver is the validated MHD2D core of the SurfaceCode pilot (HLL finite volume, MUSCL/minmod, SSP-RK2, Dedner GLM divergence cleaning, well-balanced gravity source; stratified wave propagation verified against the exact linear solution of [30] to $< 2\%$).

2 Model atmosphere — two modes

Production mode (MODE=krause, default). The background of the published Moreton-wave simulations of Krause et al. [12]: a *gravity-free, isobaric* corona ($T = 1.6$ MK, $p = 2.65 \times 10^{-3}$ Pa, $n \simeq 1.2 \times 10^8 \text{ cm}^{-3}$; gravity is negligible below ~ 100 Mm since the coronal pressure scale height is $\simeq 100$ Mm at 1.6 MK) with a dense chromospheric slab ($T = 10^4$ K, density contrast 160) below $z = 5$ Mm, joined by a resolvable smoothstep edge. This background is an *exact static solution at any resolution*, which is what makes the strong-shock/chromosphere interaction numerically robust — the reason Krause et al. adopted it.

High-resolution mode (MODE=valc). The full hydrostatic stratification, $dp/dz = -\rho g_\odot$ with constant $g_\odot = 274 \text{ m s}^{-2}$, a piecewise VAL-C temperature profile [5], and a smooth mean molecular weight $\mu(T)$ interpolating between neutral ($\mu = 1.25$) and ionized H+He ($\mu = 0.6$). Physically complete (photosphere, temperature minimum, chromosphere, TR, corona — the full Uchida waveguide), but hydrostatics fixes the chromosphere $\rightarrow$ corona density contrast at $\sim 10^5$, so the density scale length at the transition region is $\sim 0.2\text{--}0.3$ Mm for *any* resolvable $T(z)$ path; surviving the blast impact requires $\Delta z \lesssim 75$ km (a multi-hour run; the $\Delta z \geq 100$ km attempts consistently failed with TR-cell positivity losses at the flare column). Adopted anchor values (VAL III, Table 12, model C, unless noted):

3 Magnetic field and the fast-speed waveguide

Uniform, vertical $B_0 = 3$ G (default). Quiet-Sun coronal field determinations give $\sim 2\text{--}6$ G with a best value near 3 G (EIT-wave seismology, [19]); Warmuth & Mann’s global model uses an

quantity	adopted value	VAL-C / literature value	ref.
T at $\tau_{500}=1$ ($z=0$)	6420 K	6420 K (kinetic, <i>not</i> $T_{\text{eff}}=5772$ K)	[5, 6]
gas pressure at $z=0$	1.172×10^4 Pa	1.172×10^5 dyn cm $^{-2}$	[5]
temperature minimum	4170 K at $z=515$ km	4170 K at $h=515$ km	[5]
chromospheric plateau	rise to 7000 K at 2.0 Mm	6040–7160 K over 1.1–2.0 Mm	[5]
transition region	centred 2.40 Mm, width 0.25 Mm (numerically widened)	$2.5 \times 10^4 \rightarrow 4.5 \times 10^5$ K over ~ 280 km at 2.27–2.54 Mm	[5]
coronal temperature	1.5 MK, isothermal	1.4–1.6 MK quiet Sun	[10, 12]
coronal density ($z=10$ Mm)	$n_{\text{H}} \simeq 1.9 \times 10^8$ cm $^{-3}$ (hydrostatic result, not tuned)	$n_e \simeq 1.2 \times 10^8$ [12]; range $1\text{--}10 \times 10^8$ cm $^{-3}$	[12, 9]
coronal sound speed	$c_s \simeq 185$ km s $^{-1}$	180 km s $^{-1}$ at 1.4 MK	[10]
scale heights	photosph. $\simeq 150$ km; corona $\simeq 70$ Mm (emergent)	$H_\rho \simeq 130$ km; $\simeq 47$ Mm $\times T/\text{MK}$	[5, 29]

Table 1: Atmosphere. The hydrostatic integration through the adopted $T(z)$ reproduces the canonical quiet-Sun coronal base density without tuning.

effective coronal base field of 3.4 G [10]; the Krause et al. Moreton simulations bracket $B_0 = 1\text{--}10$ G uniform [12]. With $B_0 = 3$ G and the model corona, $V_A(10 \text{ Mm}) \simeq 430 \text{ km s}^{-1}$ — matching the derived literature check ($B = 3$ G, $n_e = 1.2 \times 10^8 \text{ cm}^{-3} \Rightarrow V_A \simeq 430 \text{ km s}^{-1}$) — and the ambient perpendicular fast speed is $V_f = \sqrt{c_s^2 + V_A^2} \simeq 470 \text{ km s}^{-1}$, consistent with the $\sim 300 \text{ km s}^{-1}$ base magnetosonic speed rising with height of [10] and with observed EIT-wave speeds ($170\text{--}465 \text{ km s}^{-1}$, mean 271 km s^{-1} [18]) sitting at or below it. Lateral propagation across the vertical field is a *perpendicular* fast wave; the $V_f(z)$ profile has the deep minimum ($\sim 7\text{--}10 \text{ km s}^{-1}$) around the temperature-minimum layer and the monotonic rise into the corona that the Uchida refraction mechanism requires [3]. A Moreton front at $500\text{--}2000 \text{ km s}^{-1}$ is thus Mach $\sim 5\text{--}20$ relative to the *chromospheric* fast speed — the standard argument that it must be coronally driven [9] — and Mach 1.3–3 relative to the corona.

4 Flare driver (“initial pulse”)

Impulsive-phase heating of a Gaussian low-corona volume,

$$\dot{q}(x, z, t) = q_0 \exp\left[-\frac{(x-x_{\text{fl}})^2}{2\sigma_x^2} - \frac{(z-z_{\text{fl}})^2}{2\sigma_z^2}\right] \sin(\pi t/\tau_{\text{fl}}), \quad 0 \leq t \leq \tau_{\text{fl}}, \quad (1)$$

with q_0 set so the time-integrated central heating equals $A p_0(z_{\text{fl}})/(\gamma-1)$: a peak overpressure ratio $\Delta p/p_0 \simeq A$ before expansion. The kernel is elliptical because hydrostatic *pressure* at the transition region rivals the low-corona value: a round $\sigma = 5$ Mm kernel at $z_{\text{fl}} = 10$ Mm deposits $\Delta p/p \sim 6$ *in situ* at the TR and detonates the thin layer from inside, whereas the wave physics requires the chromosphere to feel only the arriving coronal shock (Krause et al. avoid this by placing the pulse at 35 Mm; the elliptical kernel achieves the same separation in our 24 Mm box, TR deposit $\Delta p/p \lesssim 0.2$).

5 Observational targets

The run is judged against the observed phenomenology:

1. **Speeds:** first measured front velocity $845 \pm 162 \text{ km s}^{-1}$, event-mean $643 \pm 179 \text{ km s}^{-1}$ (12-event statistics [7]); historical range $500\text{--}2000 \text{ km s}^{-1}$ [1, 4].
2. **Deceleration:** mean $\bar{a} \simeq -1.5 \text{ km s}^{-2}$ (weakening with time), decaying amplitude, broadening profile — a freely decaying nonlinear fast-mode shock relaxing toward the ambient speed [7, 8].

driver parameter	adopted	literature basis	ref.
overpressure ratio A	100 (krause) / 30 (valc)	Vršnak–Cliver fiducial 20 ($M_A \simeq 1.4$), 40 for $M_A \simeq 2$; Krause et al. up to 100	[11, 12]
source size $\sigma_x \times \sigma_z$	5×2.5 Mm (elliptical)	flare-kernel $x_0 \lesssim 10$ Mm; pulse region 10 Mm in Krause et al.	[11, 12]
source height z_{fl}	10 Mm	low-corona pulse, well above the TR (35 Mm in [12])	[11, 12]
duration τ_{fl}	60 s	40–50 s pulses in [12] (40 s = min. for observable Moreton signature); impulsive phase ~ 1 –5 min	[12, 11]
temperature cap (implied)	peak $T \sim 3 \times 10^7$ K	flare-loop limits $T \leq 4 \times 10^7$ K, $n \leq 10^{11} \text{ cm}^{-3}$	[12, 29]
wave energy content	$\sim \text{few} \times 10^{20}$ J (2D slab $\times$ 30 Mm ribbon)	observed wave energies 10^{19} – 10^{21} J, i.e. ~ 0.01 –1% of the 10^{22} – 10^{25} J flare budget	[26, 9] [27]

Table 2: Flare-pulse calibration. The blast-wave route requires low- β AR-periphery conditions ($\beta_0 \sim 0.01$ –0.1) and impulsive lateral expansion [11, 16].

3. **Visibility range:** first seen at $\sim 97 \pm 26$ Mm from the source, lost at $\sim 301 \pm 47$ Mm [7].
4. **Chromospheric swing:** downward push of ~ 1 –4 km s $^{-1}$ (max 4.1 km s $^{-1}$ upper chromosphere [23]; 2.6 km s $^{-1}$ Doppler [25]), down-up relaxation over ~ 1 –3 min.
5. **Coronal counterpart:** EUV front, compression ratio 1.05–1.3 [20], SXR fast-mode Mach 1.1–1.3 [21, 22]; slower EIT fronts at $\sim 1/3$ of the coronal shock speed in the Chen et al. picture [14, 15].

6 Numerical setup and interventions

2D (x, z) slice. Krause mode: $L_x = 250$ Mm, $z \in [0, 60]$ Mm, $\Delta x = 250$ km, $\Delta z = 250$ km, pulse at $(x, z) = (40, 35)$ Mm, $A = 100$ (the observable-Moreton amplitude of [12]), $B_0 = 3$ G, $\tau_{\text{fl}} = 50$ s; H α proxy trace at the upper slab $z = 4$ Mm (where the skirt swing is largest), coronal trace at $z = 15$ Mm. VAL-C mode: $L_x = 200$ Mm, $z \in [-0.6, 24]$ Mm, $\Delta x \leq 250$ km, $\Delta z \lesssim 75$ km; flare centred at $x_{\text{fl}} = 30$ Mm; mass-conserving absorbing layers (velocity + internal-energy relaxation toward the hydrostatic background) at the top and both lateral boundaries. Diagnostics: 2D snapshots of v_z , v_x , $\Delta p/p_0$ every 30 s; distance–time stacks (1 s cadence) of chromospheric v_z at $z = 1.5$ Mm (H α proxy) and coronal $\Delta p/p_0$ at $z = 8$ Mm (EUV proxy); front tracking and kinematics.

Known simplifications. (i) Ideal MHD: no radiative losses, thermal conduction, or partial-ionization physics — acceptable on the minutes-long wave-crossing time, but it leaves the *flare site itself* with no physical relaxation channel. We therefore gently relax the velocity in a Gaussian column around the flare site after the pulse ends ($t > \tau_{\text{fl}}$, rate 0.1 s $^{-1}$, $\sigma_x = 2\sigma_{\text{fl}}$, $z \lesssim 10$ Mm): in ideal MHD the post-flare cavity refill otherwise steepens indefinitely against the numerically thin transition region. The escaping front is $\gtrsim V_f \tau_{\text{fl}} \sim 30$ Mm away when the relaxer activates; front kinematics are read at $x - x_{\text{fl}} \gtrsim 30$ Mm. (ii) The flare heating is applied along the whole column (its low-atmosphere share is small in relative terms, $\Delta p/p \lesssim 0.3$ below the TR, and mimics ribbon co-heating); test runs that artificially confined the heating to the corona created an unphysical razor pressure interface on the transition region. (iii) Uniform vertical field: no arcade geometry, no field-line-stretching (pseudo-wave) component — this run isolates the *wave* part of the EIT-wave duality [14, 9]. (iv) 2D slab: no cylindrical geometric spreading, so amplitude decay along x is a conservative (upper) limit on front survival. (v) A positivity backstop: the HLL scheme is not positivity-preserving, and blast-driven rarefactions can empty a (numerically thin) transition-region cell; cells falling below $10^{-3}\rho_0(z)$ are floored with their momentum zeroed (physical wake

evacuation stays $\gtrsim 10^{-1}\rho_0$, so the backstop only catches numerical failures; hit counts are logged).
(vi) The $\text{H}\alpha$ signature is represented by the chromospheric velocity swing, not synthesized radiative transfer (an 8% optically-thin compression threshold is the visibility criterion used by [12]).

7 Results: comparison with the observed 2006 December 6 Moreton wave

The production run (MODE=krause, $A = 100$, $B_0 = 3$ G, $\Delta x = \Delta z = 250$ km) is compared against the best-observed recent Moreton wave, the 2006 December 6 event (3B/X6.5 flare in AR 10930), analysed by Balasubramaniam et al. [24] (RefMoreton/) — the same event modelled by Krause et al. [12]. Front tracking mirrors their method: the leading edge is followed and fitted with the constant-deceleration model $d = d_0 + V_0 t + \frac{1}{2}at^2$ (their Eq. 1). Figure [plots_production/fig5_obs_comparison.png](#):

	simulation	observed [24]
mean front speed	857 km s ⁻¹ (41–226 s)	~ 850 km s ⁻¹ (18:43–18:50 UT)
constant-deceleration fit V_0	891 km s ⁻¹	~ 1125 km s ⁻¹
deceleration a	-0.42 km s ⁻²	-0.87 km s ⁻²
final tracked speed	813 km s ⁻¹	~ 700 – 800 km s ⁻¹ (approaching ambient)
chromospheric swing amplitude	2.0 km s ⁻¹ (far field)	1–4 km s ⁻¹ (typical events [23, 25])
chromospheric signal range	31–126 Mm from source	~ 120 – 550 Mm from radiant point
coronal front tracked to	204 Mm ($\Delta p/p < 3\%$)	EUV counterpart to $\gtrsim R_\odot$ scales

Table 3: Simulated vs. observed kinematics of the 2006 December 6 Moreton wave.

Agreement. The propagation-phase kinematics — the observable Moreton signature — are reproduced: the mean front speed matches the observed 850 km s⁻¹ to within the measurement error, the front decelerates at the same order (-0.4 vs. -0.9 km s⁻², both fits weakening with time), the final speed approaches the ambient fast-mode speed from above (the freely decaying weak-shock behaviour of [8, ?]), and the chromospheric down-up swing amplitude falls inside the observed 1–4 km s⁻¹ band.

Differences, and why. (i) The observed wave stayed traceable to ~ 550 Mm; our chromospheric signal drops below the 0.3 km s⁻¹ threshold at ~ 126 Mm and the coronal front below $\Delta p/p = 3\%$ at ~ 204 Mm (the domain ends 210 Mm from the source). The 2006 event was extreme (X6.5, $\sim 270^\circ$ azimuthal span, the strongest field change of ~ 80 surveyed flares), our pulse is the *minimum observable-Moreton* amplitude of [12], and a 2D slab under-represents the sustained lateral driving of a real, volumetric CME expansion — all three push the simulated visibility range low. (ii) The observed wave was first detected only at ~ 120 Mm because flare emission saturates $\text{H}\alpha$ closer in; our inner cutoff (31 Mm) is the flare-column guard zone — analogous in effect, not in cause. (iii) V_0 is $\sim 20\%$ low: the fit begins at our first clean detection ($t = 41$ s), missing the strongly super-magnetosonic launch phase the observed back-extrapolation captures. *Driver caveat:* [24] conclude that for this specific event the kinematics favour lateral CME-arcade expansion over a pure flare pressure pulse (the pulse would need ~ 100 km s⁻² acceleration and -10 km s⁻² deceleration to fit their back-extrapolated launch). Our blast-driven run cannot arbitrate the launch mechanism — but once launched, the propagating fast-mode front forgets its driver, which is exactly what the agreement in the propagation phase demonstrates.

8 Referee test: driver-limited vs. dissipation-limited range

A natural objection: if the impulsive run’s front fades by ~ 200 Mm while the real Moreton wave was seen to ~ 550 Mm, perhaps the solver simply *over-dissipates* — in which case the T Tauri

decay-length result (the parent paper) would be suspect for the same reason. This is testable directly. If the short range is set by the *driver* (a weak, impulsive pulse in a finite box) rather than by *dissipation*, then driving the wave harder and for longer, in a larger domain, at *identical* solver, dissipation physics and resolution, should extend the range toward the observed value.

We therefore added a sustained CME-piston driver (**DRIVE=piston**): steady energy deposition at rate $A p_0 / [(\gamma - 1) \tau_{\text{fl}}]$ held over $\tau_{\text{drive}} = 180$ s (the CME lateral-acceleration phase, cosine-ramped), with the driven lateral extent growing as $\sigma_x(t) = \sigma_{\text{fl}} + V_{\text{pist}} t$ ($V_{\text{pist}} = 800 \text{ km s}^{-1}$) so the driving edge tracks outward with the front — the continuous lateral push of an expanding CME flank (Chen et al. [14]; Vrřnak & Cliver [11]), as opposed to the impulsive blast. Domain $L_x = 600$ Mm, $A = 150$, run to $t = 720$ s.

run (same solver, physics, $\Delta x = \Delta z = 250$ km)	mean speed	front reached	driver
impulsive blast ($A=100$, 210 Mm box)	857 km s^{-1}	204 Mm	one-shot
sustained CME-piston ($A=150$, 600 Mm box)	833 km s^{-1}	553 Mm	continuous
observed 2006 December 6 [24]	$\sim 850 \text{ km s}^{-1}$	~ 550 Mm	CME (inferred)

Table 4: The visibility range is set by the driver, not by numerical dissipation. Figure `plots_extreme/fig6_driver_test.png`.

The sustained-driver front reaches **553 Mm** — essentially the observed ~ 550 Mm — while propagating at the same $\sim 830\text{--}850 \text{ km s}^{-1}$ mean speed, and its distance–time track brackets the observed curve across the whole disk. Nothing about the dissipation changed between the two runs; only the driver and the box did. *The short range of the impulsive run is driver-limited, not dissipation-limited*, which is the premise the objection needs and does not have.

Honest cost of the extreme driver. The sustained $A = 150$ piston over-drives the chromospheric response: the far-field vertical swing reaches $\sim 130 \text{ km s}^{-1}$, far above the observed $1\text{--}4 \text{ km s}^{-1}$ (the impulsive run gave a realistic 2 km s^{-1}). So the two runs bracket reality — the impulsive blast reproduces the *amplitude* but under-ranges; the extreme piston reproduces the *range* but over-drives the amplitude — and a real, moderate CME sits between. Neither the deceleration nor V_0 of the piston run should be read as free-decay values: its fit is contaminated by the 180 s driven phase (hence the inflated $V_0 \sim 2500 \text{ km s}^{-1}$ and $a \sim -6 \text{ km s}^{-2}$). The decisive, driver-independent quantities are the mean propagation speed and the range, both of which match.

Bearing on the T Tauri result. The parent paper’s decay length does not rest on trusting the scheme’s dissipation in the first place: the numerical dissipation there was *measured* (the dedicated numerical-diffusion test), and at the converged resolution the numerical decay length is $\sim 100\times$ longer than the reported physical $L \simeq 0.006 R_\star$, with L drifting only $+15\%$ over a $\Delta x = 50 \rightarrow 13$ km refinement ladder (converging, not grid-limited). Moreover the two decays are different physics — a Mach-3 shock plus *ambipolar* friction in near-neutral gas (T Tauri) vs. a weakly nonlinear ideal-MHD wave (solar corona) — so there is no shared dissipation knob whose miscalibration would inflate both. The present solar test removes even the indirect worry: when the solver is asked to carry a fast-mode front ~ 550 Mm at the correct speed, it does.

8.1 Can a single driver hit both the range and the amplitude? A trade-off scan

Pushing further: the extreme run reproduces the *range* but over-drives the chromospheric swing ($\sim 130 \text{ km s}^{-1}$ vs. observed $1\text{--}4 \text{ km s}^{-1}$). Can an intermediate sustained driver hit both? We scanned the piston amplitude ($A = 20, 25, 50, 150$; the $A = 25$ case also tripled the driver width, $\sigma_x = 15$ Mm, to test a “wide-and-gentle” CME):

Two robust facts emerge. (i) The *range* saturates at ~ 550 Mm for any $A \gtrsim 25$ — it is set by sustained driving and the detection threshold, not by amplitude, and it does *not* increase further with a stronger driver. (ii) The *swing* grows with the total deposited energy; widening the driver at

run	A (width)	mean speed	range ($\Delta p/p > 3\%$)	far-field swing
mid20	20	726 km s^{-1}	439 Mm	27 km s^{-1}
widegentle	25 ($\sigma_x=15$)	802 km s^{-1}	553 Mm	78 km s^{-1}
mid50	50	778 km s^{-1}	548 Mm	63 km s^{-1}
extreme	150	831 km s^{-1}	553 Mm	133 km s^{-1}
<i>observed</i> [24]	—	$\sim 850 \text{ km s}^{-1}$	$\sim 550 \text{ Mm}$	$1\text{--}4 \text{ km s}^{-1}$

Table 5: Range vs. chromospheric-swing trade-off across the sustained-driver scan (Figure `plots.tradeoff/fig7_tradeoff.png`). Every run that reaches the observed ~ 550 Mm range has a swing $15\text{--}30\times$ above the observed band; the “sweet-spot” corner (range ~ 550 Mm *and* swing $1\text{--}4 \text{ km s}^{-1}$) is empty.

fixed A (`widegentle`) made it *worse*, not better, because a wider Gaussian deposits more energy. So a single sustained-driver knob cannot reach the observed corner.

Why the real Sun hits both, and why this strengthens (not weakens) the argument.

The missing ingredient is *geometric spreading*, which the 2D slab omits by construction (§5, simplification iv). On the real Sun the front expands as an arc, so its *local* amplitude falls as $\sim r^{-1/2}$ (cylindrical) *in addition* to dissipation; a front seen at 550 Mm is therefore intrinsically weak there — large range *and* small local swing. Our 2D slab has no such spreading, so the only way it can keep a front detectable to 550 Mm is to keep it strong, forcing a large swing. The trade-off is thus an artifact of the deliberately 2D geometry, in the *conservative* direction: with spreading restored, the same front would decay *faster* in amplitude, not slower. This is the same reason the paper flags the 2D decay lengths as *upper limits* — and it is the opposite of the “code over-dissipates” worry. The dissipation is, if anything, under-represented here; the wave still crosses the disk because it is driven, not because the scheme conserves its energy artificially.

9 Relation to the T Tauri SurfaceCode pilot

`SolarRuns` reuses the `SurfaceCode` solver core unchanged, but the two applications sit at opposite ends of the same physical family: both simulate an impulsively driven fast-magnetosonic front sweeping laterally away from a localized surface source on a magnetized star — yet in *different atmospheric channels*, and that difference is the physics.

The two results are mutually consistent rather than contradictory: lateral fast-wave transport survives only in the tenuous exterior (solar corona), not in the dense surface layers (T Tauri photosphere) — and even the dense-layer footprints agree quantitatively, since the T Tauri decay length (~ 8 Mm) and the distance over which the simulated solar *chromospheric* response stays strong ($\lesssim 100$ Mm of the source, §5) are both small compared to the stellar circumference. This is precisely the front-vs-energy dichotomy of the parent paper: a coronal front can cross the disk while the dense-layer energy density it drags along fades within a fraction of a radius. The solar run is therefore both a demonstration that the `SurfaceCode` core handles the classic Moreton problem, and an independent illustration of why the accretion-powered surface channel on T Tauri stars — which has no hot corona to ride — cannot deliver launch-capable wave energy to the pole.

References

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	T Tauri pilot (parent)	SolarRuns (this work)
question	does the accretion-impact wave survive ring→pole? ($T = F_{\text{pole}}/F_{\text{ring}}$)	does the flare blast reproduce the observed Moreton phenomenology?
star	$0.5 M_{\odot}$, $2 R_{\odot}$ T Tauri, $g \simeq 34 \text{ m s}^{-2}$	the Sun, $g_{\odot} = 274 \text{ m s}^{-2}$
propagation channel	<i>dense</i> photospheric layers ($\tau \gtrsim 1$ to $\tau \ll 1$), the layer Cranmer’s transport assumes	<i>tenuous</i> corona; the chromosphere only <i>prints</i> the front (Uchida skirt)
background	real MLT envelope (Saha EOS export), hydrostatic, isothermal placeholder above $\tau=1$	Krause slab mode (gravity-free, isobaric) or VAL-C hydrostatic mode; §2
magnetic field	ring-latitude dipole, 0.1–1 kG (kilogauss star)	uniform vertical quiet-Sun 3 G
driver	accretion-clump impact: boundary velocity piston, $\sim \text{Mach } 3$ at the photosphere, $\tau = 150 \text{ s}$	flare pressure pulse: thermal overpressure $\Delta p/p_0 = 50$ in the low corona, $\tau_{\text{ff}} = 50 \text{ s}$
key damping physics	nonlinear shock + <i>ambipolar friction</i> (neutral photosphere; the T-Tauri-specific term)	ideal MHD (fully ionized corona; shock dissipation only)
resolution	$\Delta x = 50 \text{ km}$ ($H_p \simeq 700 \text{ km}$)	$\Delta x = \Delta z = 250 \text{ km}$ (Mm-scale structures; finer than [12])
new numerics	— (validated core)	per-stage bounded-state protection, flare-column guard/relaxer zones, gravity-free isobaric mode
diagnostic	transmitted flux $\Phi(x)$, decay length L	front kinematics $d(t)$, $v(t)$; chromospheric down-up swing; EUV-proxy front
outcome	the dense channel <i>kills</i> the wave: $L \simeq 0.006 R_{\star}$ ($\sim 8 \text{ Mm}$), buried in ambient convection	the coronal channel <i>carries</i> the front across $\gtrsim 100\text{--}200 \text{ Mm}$ at $\sim 500\text{--}800 \text{ km s}^{-1}$, decelerating

Table 6: The same solver, the two opposite transport channels.

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